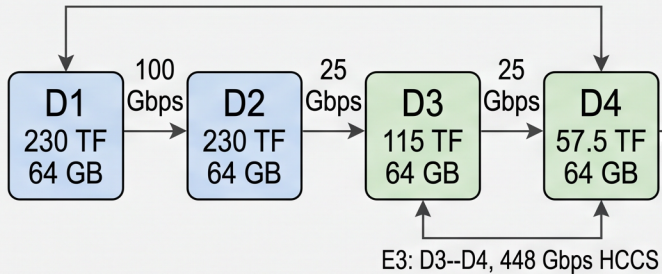
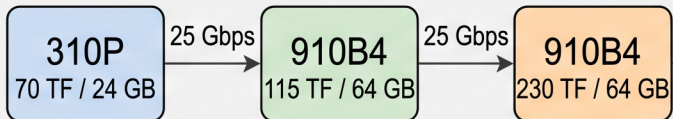


Hardware Topology H=(V,E)

E1 / E2



E4



Model and Workload

Model Profile

CodeLlama34B:
48 Text Layers, 67.5 GB
Qwen2-VL-7B: Vision Encoder + Projector + 28 Text Layers

Layer FLOPs
Weights
Activation
KV Cache

Workload and SLA

Input Length
Output Length
Visual Tokens
Micro-batches
TTFT
TPOT
E2E

Unified Cost Model

Stage Compute
Memory Feasibility

TP Collective Communication

PP Boundary Transfer

Pipeline Fill and Bubble

Deployment Alternatives

TP

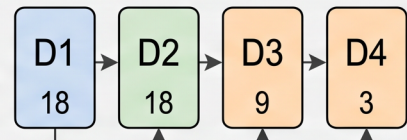
Connected Device Group

D1 + D2

Tensor Shards

Collective Communication per Layer

PP



PP+TP

